Akshat Choube

choube.a@northeastern.edu
+1 (213) 800-6697
https://akshat3011.github.io/

My overarching research interest lies in developing Machine Learning and Artificial Intelligence systems that are "behavior-aware," capable of understanding users on a deeper level and supporting more personalized experiences. Currently, I focus on designing human-centered, multi-agent AI and ML systems that use passive sensing data from phones and wearables to understand behavioral patterns and generate meaningful health & well-being insights. My goal is to personalize these insights so that it is accessible and actionable to a wide range of stakeholders, including behavioral researchers, psychologists, and individuals interested in health and well-being.

Education

2023 - Present **Ph.D., Computer Science**
Northeastern University
Advisor: Varun Mishra

2019 - 2021 **MS (Hons), Computer Science**
University of Southern California, Los Angeles

2015 - 2019 **BTech, Computer Science & Engineering**
Indian Institute of Technology (IIT), Palakkad

Experience

Sept 2026 - Dec 2026 **Visiting Researcher**, Università della Svizzera italiana (USI)
Working with Prof. Silvia Santini in the People-Centered Computing Lab on research at the intersection of ubiquitous computing, mobile sensing, and human-centered AI.

June 2026 - August 2026 **Ph.D. AI Research Intern**, Optum Labs, UHG
Developed a multi-agentic system to automate post-training of LLMs in clinical settings, including automatic discovery of novel methods, implementation, and evaluation.

Sept 2023 - Present **Graduate Research Assistant**, UbiWell Lab, Northeastern University
Developing AI and ML systems that transform passive sensing data into actionable, accessible insights for health and wellbeing.

Jul 2021 - Jul 2023 **Software Development Engineer (ML)**, Amazon Search
Worked with Query Understanding and Autocomplete teams. Developed ML and NLP solutions to better understand shopping intent on Amazon, improving query parsing, intent detection, to show relevant products and query suggestions.

Jan 2021 - May 2021 **Graduate Research Assistant**, Intelligent Human Perception Lab, USC
Built models to predict self-reported mood using Apple Watch data (such as heart rate, physical activity, etc.). Applied statistical analysis and deep learning to uncover correlations between physiological signals and mood.

Jan 2021 - May 2021 **Teaching Assistant** , Viterbi School of Engineering, USC
Held office hours to help students with NLP concepts, project development, and report writing. Assisted with grading and facilitated project brainstorming sessions. Presented my paper on multimodal machine learning to a class of 150 students.

Jun 2020 - Aug 2020 **Software Development Engineer Intern (ML)**, Amazon Search, Palo Alto
Designed and implemented a Spark-based data processing framework to convert huge offline user behavior and clickstream data into a read-only database (RODB) for deployment in online search services. Enabled customized preprocessing (filtering, transformation, aggregation, merging) for datasets before conversion.

Sep 2019 - Jan 2021 **Graduate Research Assistant**, Social Media Analytics Lab, USC
Applied NLP to analyze public health behaviors on Twitter, focusing on conversations, marketing, and misinformation related to tobacco and cannabis products. Investigated the role of social bots in promoting harmful content on the platform, especially looking at at-risk populations (such as young adults and pregnant women).

May 2016 - Sept 2016 **Junior Research Fellow**, Tata Institute of Fundamental Research, Mumbai
Studied information exchange system involving bit packets; defined a novel Quality of Service (QoS) metric based on transmission time and packet fraction. Proved optimal transmission ordering and designed a greedy algorithm with a guaranteed 2-approximation to the intractable optimal solution.

Publications

[C1] **Choube**, **Akshat**, Le, Ha, Li, Jiachen, Fu, Shihan, Watanabe, Yuna, Intille, Stephen, and Mishra, Varun. "Tutorial: Trials and Tribulations of Making Multi-Agent AI Systems with Sensing Data Work: What You Need to Know". In: New York, NY, USA: Association for Computing Machinery, 2026, pp. 1–4. DOI: 10.1145/3798063.3836757.

[C2] **Choube**, **Akshat**, Sharifi Renani, Mohsen, Ghaffari, Meysam, Wang, Junda, Mishra, Varun, and Morato, Carlos. "AutoCon: A Multi-Agent System for Autonomous Clinical Model Post-Training". In: Under submission (2026).

[C3] Kim, Jin-seo, Le, Ha, **Choube**, **Akshat**, Mishra, Varun, and Intille, Stephen. "Generating Personalized Games with Sensing Data for Data Collection". In: UIST '26. New York, NY, USA: ACM, 2026, pp. 11. DOI: 10.1145/3830398.3830678.

[C4] **Choube**, **Akshat**, Shrestha, Shreeti, Le, Ha, Li, Jiachen, Das Swain, Vedant, and Mishra, Varun. "DAIMON: Designing AI-Augmented Research Dashboards to Enable Novel Human-AI Collaborative Workflows in Longitudinal Sensing Studies". In: Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 10.3 (2026). DOI: 10.1145/3832033.

*: Equal Contributions

[C5] Le, Ha, **Choube**, **Akshat**, Mishra, Varun, and Intille, Stephen. "Feasibility of using a Multi-Agent LLM System to Correct Annotations and Support Low-Effort Activity Labeling". In: Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 10.3 (2026). DOI: 10.1145/3831653.

[C6] **Choube**, **Akshat**, Le, Ha, Li, Jiachen, Ji, Kaixin, Swain, Vedant Das, and Mishra, Varun. "GLOSS: Group of LLMs for Open-Ended Sensemaking of Passive Sensing Data for Health and Wellbeing". In: Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT) 9.3 (2025), pp. 1–32.

[C7] Le, Ha*, **Choube**, **Akshat***, Mishra, Varun, and Intille, Stephen. "A Multi-Agent LLM Network for Suggesting and Correcting Human Activity and Posture Annotations". In: *Companion of the 2025 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*. 2025, pp. 6. DOI: 10.1145/3714394.3756185.

[C8] Le, Ha, Potter, Veronika, **Choube**, **Akshat**, Lakshminarayanan, Rithika, Mishra, Varun, and Intille, Stephen. "A Context-Assisted, Semi-Automated Activity Recall Interface Allowing Uncertainty". In: Proceedings of the ACM on interactive, mobile, wearable and ubiquitous technologies 9.4 (2025), pp. 1–34.

[C9] **Choube**, **Akshat**, Bhattacharya, Sohini, Majethia, Rahul, Li, Jiachen, Das Swain, Vedant, and Mishra, Varun. "Imputation Matters: A Deeper Look into an Overlooked Step in Longitudinal Health and Behavior Sensing Research". In: Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 9.4 (2025). DOI: 10.1145/3770647.

[C10] Bhattacharya, Sohini*, Majethia, Rahul*, **Choube**, **Akshat***, and Mishra, Varun. "Imputation Strategies for Longitudinal Behavioral Studies: Predicting Depression Using GLOBEM Datasets". In: *Companion of the 2024 on ACM International Joint Conference on Pervasive and Ubiquitous Computing*. 2024, pp. 736–742.

[C11] **Choube**, **Akshat**, Swain, Vedant Das, and Mishra, Varun. "SeSaMe: A Framework to Simulate Self-Reported Ground Truth for Mental Health Sensing Studies". In: *2024 12th International Conference on Affective Computing and Intelligent Interaction (ACII)*. IEEE. 2024, pp. 228–237.

[C12] Allem, Jon-Patrick, Dormanesh, Allison, Majmundar, Anuja, Unger, Jennifer B, Kirkpatrick, Matthew G, **Choube**, **Akshat**, Aithal, Aneesh, Ferrara, Emilio, and Boley Cruz, Tess. "Topics of nicotine-related discussions on Twitter: infoveillance study". In: Journal of medical Internet research 23.6 (2021), pp. e25579.

[C13] **Choube**, **Akshat** and Soleymani, Mohammad. "Punchline detection using context-aware hierarchical multimodal fusion". In: *Proceedings of the 2020 International Conference on Multimodal Interaction (ICMI)*. 2020, pp. 675–679.

[C14] Vaze, Rahul, Chaudhari, Shreyas, **Choube**, **Akshat**, and Aggarwal, Nitin. "Energy-delay-distortion problem". In: *2018 Twenty Fourth National Conference on Communications (NCC)*. IEEE. 2018, pp. 1–6.

Invited Talks

2026 *Reducing barriers in collecting and generating insights from longitudinal digital phenotyping data for health and wellbeing*
Center for Technology and Behavioral Health, Dartmouth College

2026 *Reducing barriers in collecting and generating insights from longitudinal digital phenotyping data for health and wellbeing*
Social Development and Wellbeing Lab, Northeastern University

Scholarships

ThinkSwiss 2026: Swiss Government–supported research scholarship for a visiting research appointment at Università della Svizzera italiana (USI), Switzerland.

Service

Reviewing

PACM CHI (2025, 2026)
PACM IMWUT (2024, 2025, 2026)
PACM MobileHCI (2026)
IEEE TAFCC (2025)

Mentoring

- **Masters Students:**
  Northeastern: Munan Han, Darsh Pandya, Shreeti Shrestha
- **Undergraduate Students:**
  Northeastern: Win Tongtawee, Arihant Pant, Sohini Bhattacharya